广州联网科技

信号接收强度说明书

D-DWM-PG1.7/2.5/3.6/PLUS/2.3/4.6/4.9

信号接收强度说明书

竭诚感谢您使用本公司的产品

本手册就产品的使用方法与安全事项进行说明

*熟读本手册，并在使用过程中注意安全。

*保留本手册，放在合适的地方以便随时查阅。

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安全注意

阅读本手册后，请妥善保管以便查阅。

这里展示的是注意事项和安全相关的重大内容，所以请一定要遵守，标志意思如下：

警 告 在操作时违反本警告事项所示的内容，可能会导致人员死亡或重伤。

注 意 在操作时违反本注意事项所示的内容，可能会导致人员负伤或造成物品损坏。

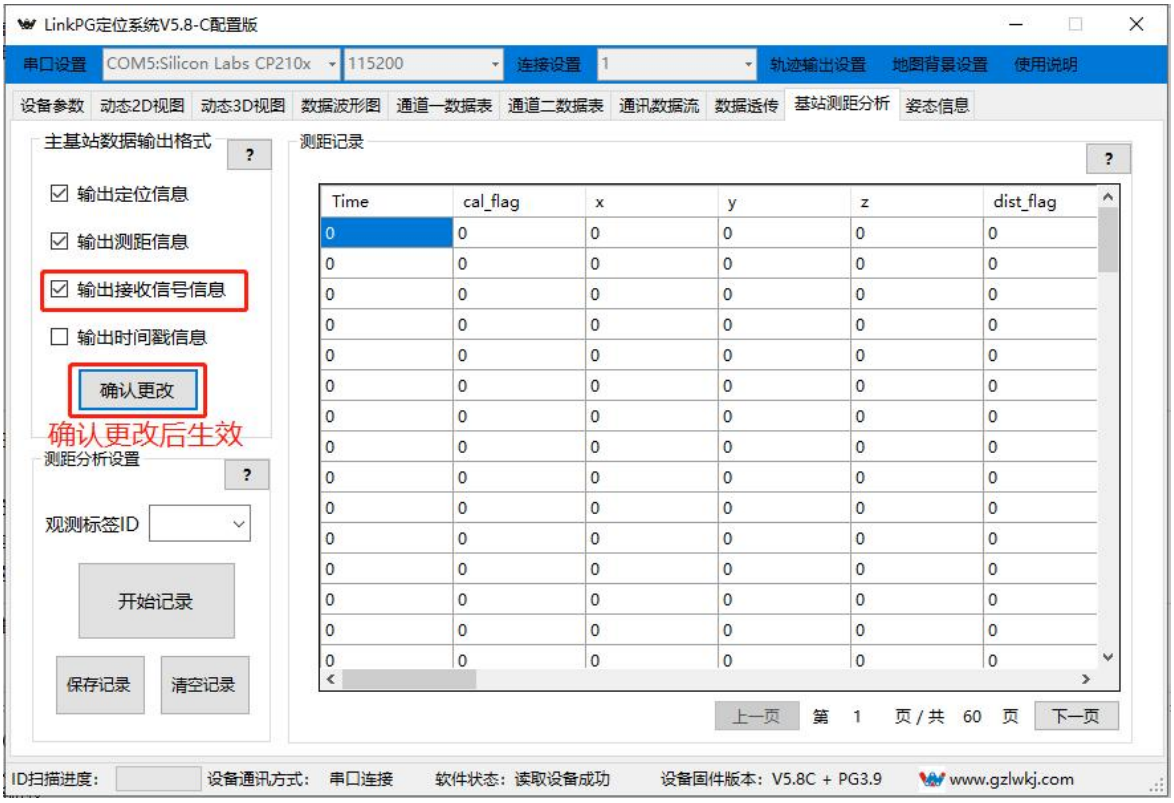
提 醒 在操作时使您能正确使用产品时，所务必遵守的相关使用

1、说明

本文档是对 PG 系统上位机基站接收强度参数列表的说明，数据说明均来自 Decawave 官方文档，程序是依据官方数据手册将信号参数提取出来供用户研究分析，原理性信号含义需用户查阅更多官方手册以及学习无线信号的知识进行深入解析。

2、信号接收强度参数说明

接收信号强度可以通过 A 基站中使能输出接收信号信息来让自动上报测距数据中包含接收信号强度信息。



系统开始定位后，选择需要记录数据的标签 ID 并点击“开始记录”。系统会自动记录该标签测距的接收信号强度相关参数到表格中。后续可以通过“保存记录”将数据保存到本地 excel 表格。

对参数列表的说明：

Fields	Description of fields within the *dwt_rxdiag_t* structure
maxNoise	The *maxNoise* parameter.
firstPathAmp1	First path amplitude is a 16-bit value reporting the magnitude of the leading edge signal seen in the accumulator data memory during the LDE algorithm's analysis. The amplitude of the sample reported in this *firstPathAmp* parameter is the value of the accumulator tap at index given by floor(*firstPath)* reported below. This amplitude value can be used in assessing the quality of the received signal and/or the receive timestamp produced by the LDE.
firstPathAmp2	Is a 16-bit value reporting the magnitude of signal at index floor (*firstPath)* +2.
firstPathAmp3	Is a 16-bit value reporting the magnitude of signal at index floor (*firstPath)* + 3.
stdNoise	The *stdNoise* parameter is a 16-bit value reporting the standard deviation of the noise level seen during the LDE algorithm's analysis of the accumulator data. This value can be used in assessing the quality of the received signal and/or the receive timestamp produced by the LDE.
maxGrowthCIR	Channel impulse response max growth is a 16-bit value reporting a growth factor for the accumulator which is related to the receive signal power. This value can be used in assessing the quality of the received signal and/or the receive timestamp produced by the LDE.
rxPreamCount	This reports the number of symbols of preamble accumulated. This may be used to estimate the length of TX preamble received and also during diagnostics as an aid to interpreting the accumulator data. It is possible for this count to be a little larger than the transmitted preamble length, because of very early detection of preamble and because the accumulation count may include accumulation that continues through the SFD (until the SFD is detected).

Fields	Description of fields within the *dwt_rxdiag_t* structure
firstPath	First path index is a 16-bit value reporting the position within the accumulator that the LDE algorithm has determined to be the first path. This value is set during the LDE algorithm's analysis of the accumulator data. This value may be of use during diagnostic graphing of the accumulator data, and may also be of use in assessing the quality of the received message and/or the receive timestamp produced by the LDE. The first path (or leading edge) is a sub-nanosecond quantity. Each tap in the accumulator corresponds to a sample time, which is roughly 1 nanosecond (or 30 cm in terms of the radio signal's flight time through air). To report the position of the leading edge more accurately than this 1-nanosecond step size, the index value consist of a whole part and a fraction part. The 16-bits of *firstPath* are arranged in a fixed point "10.6" style value where the low 6 bits are the fractional part and the high 10 bits are the integer part. Essentially this means if the *firstPath* is read as a whole number, then it has to be divided by 64 to get the fractional representation.

具体的信息可查看"DW1000_Software_API_Guide_rev2p7"文档 5.60 部分的内容。

右侧接收强度计算说明：（user manual 的 4.7.1）

4.7.1 Estimating the signal power in the first path

An estimate of the power in the first path signal may be calculated (in dBm) using the formula:

$$First\ Path\ Power\ Level = 10 \times \log_{10}\left(\frac{F_1{}^2 + F_2{}^2 + F_3{}^2}{N^2}\right) - A\ \ dBm$$

Where:

F1 = the *First Path Amplitude (point 1)* magnitude value reported in the FP_AMPL1 field of *Register file: 0x15 – Receive Time Stamp*,

F2 = the *First Path Amplitude (point 2)* magnitude value reported in the FP_AMPL2 field of *Register file: 0x12 – Rx Frame Quality Information*,

F3 = the *First Path Amplitude (point 3)* magnitude value reported in the FP_AMPL3 field of *Register file: 0x12 – Rx Frame Quality Information*,

A= is the constant 113.77 for a PRF of 16 MHz, or, the constant 121.74 for a PRF of 64 MHz, and

N = the *Preamble Accumulation Count* value reported in the RXPACC field of *Register file: 0x10 – RX Frame Information Register*. Note that RXPACC may need to be adjusted to remove SFD symbol count before use, see the register field description in *Register file: 0x10 – RX Frame Information Register*.

The resultant *First Path Power Level* (in dBm) may be compared with the estimated receive power figure calculated as per section *4.7.2 – Estimating the receive signal power*.

这里 N 使用的是 rxPreamCount 的值

4.7.2 Estimating the receive signal power

It is possible to calculate an estimate of the receive power level (in dBm) using the formula:

$$RX\ Level = 10 \times \log_{10}\left(\frac{C \times 2^{17}}{N^2}\right) - A\ \ dBm$$

Where:

C = the *Channel Impulse Response Power* value reported in the CIR_PWR field of *Register file: 0x12 – Rx Frame Quality Information*,

A= is the constant 113.77 for a PRF of 16 MHz, or, the constant 121.74 for a PRF of 64 MHz, and

N = the *Preamble Accumulation Count* value reported in the RXPACC field of *Register file: 0x10 – RX Frame Information Register*. Note that RXPACC may need to be adjusted to remove SFD symbol count before use, see the register field description in *Register file: 0x10 – RX Frame Information Register*.

This resultant receive power estimate is very close to the actual receive power at lower receive levels, but is lower than the actual receive power level at higher levels. Figure 22 below shows the relationship between the actual receive power and the power estimated by this technique.

这里 C 是使用 maxGrowthCIR 的数值。

更详细的信息可查看数据手册“DW1000 User Manual_2.18 ”的 4.7.1 部分内容。

更新日志			
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